Waipapa Taumata Rau
University of Auckland

Part IV Project #123
Department of Electrical, Computer, and Software Engineering

AI-Based Virtual Parliament

Cooperative debate and hidden disruptive agents

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Real MPs can derail debate, distort claims and exploit procedure. AI multi-agent debate has been studied to improve reasoning [1]. We built a virtual NZ House of AI MPs, then privately tasked some with sabotage and tested the effects of grounding agents with Hansard.

1 Parliament web browser interface

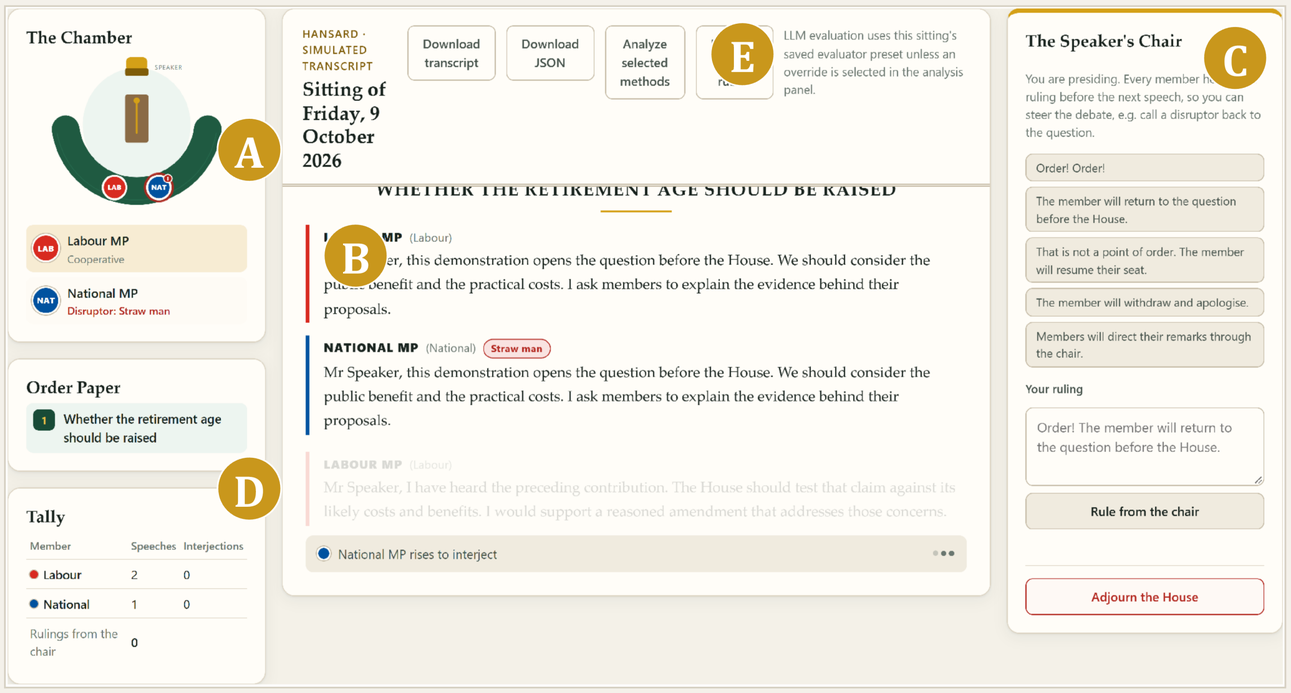


Fig. 1 A demonstration sitting with a National straw-man assigned adversarial agent. Web browser, guided command, and terminal interfaces all connect to the debate engine backend.

A The Chamber
One MP represents each chosen party.
Speakers are highlighted when speaking.

B Live Hansard
Speeches and interjections stream into the live transcript.
Saved sittings can be replayed after a restart.

C The Speaker's Chair
Chair rulings enter the shared public record and reach every MP before the next speech.

D Order Paper and Tally
The agenda advances through topics in order.
Live totals track speeches, interjections and rulings.

E Exports and evaluation
Public JSON and text exports support local sentiment/stance analysis and separate rubric scores.

2 Cooperative and adversarial agents

At run start, the backend combines each MP's party profile and selected real parliamentary debate excerpts [2] in a private system prompt, adding any assigned tactic. The scheduler takes turns sending that MP's prompt, public debate history and a turn cue to its model's chat completion endpoint. Each returned speech joins the public record for the next MP. Other agents cannot read the private prompts.

Real Hansard
500 reviewed excerpts
0/10/100 per party

→ **Private setup**
Frozen prompts
and hidden tactics

→ **MP agents**
Party personas
on cloud or local LLMs

→ **Debate engine**
Scheduled turns
and chat requests

→ **Saved evidence**
Public transcript
and separate reports

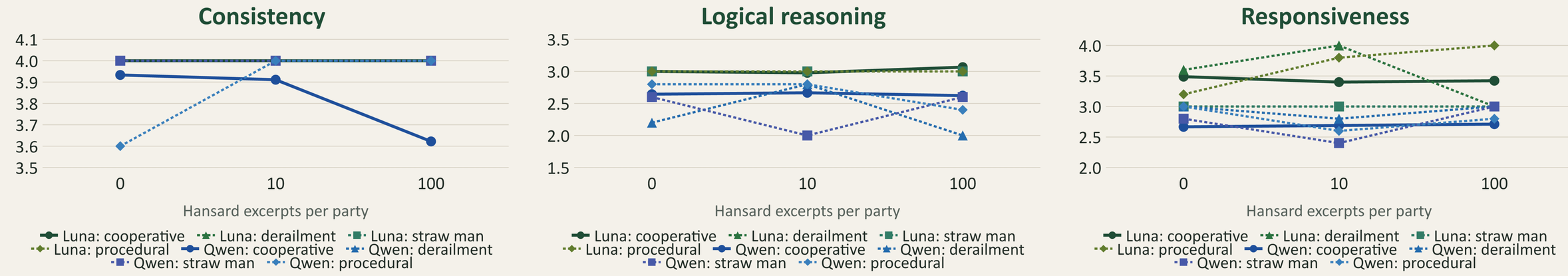
Topic derailment
Steer discussion away from the agenda without acknowledging it.

Straw man
Misrepresent an opponent's claim, then attack the distorted version.

Procedural manipulation
Exploit rules or points of order to obstruct substantive debate.

3 Grounding and adversarial testing

72 debates (1,800 speeches): **54 cooperative baselines** (three repeats), plus **18 adversarial runs** (one repeat, one assigned MP). GPT-6 Luna and Qwen3:8b debate retirement, nuclear power and a four-day workweek at 0/10/100 excerpts, with five parties, five rounds and no interjections.



Vertical axes are truncated. Rubric scores range from 0 to 4. Each co-operative point averages 9 debates (45 MP ratings per metric); each strategy point averages five MPs. GPT-6 Luna used a consistent rating prompt. Topic, party and repeats limit comparison.

Co-operative agents: Consistency stays high, but Qwen falls from 3.93 to 3.62. Luna's reasoning stays near 3.0; Qwen's stays between 2.62 and 2.67. Responsiveness changes little for either model. More Hansard gives no consistent gain.

Adversarial agents: Qwen's reasoning varies from 2.0 to 2.8; Luna's procedural responsiveness rises from 3.2 to 4.0. Disruption scores are 2/4 for Qwen ACT only at 0 excerpts and 0 for adversarial Luna MPs; Qwen Green's paired changes span -1 to +1. Adversarial assignment alone is insufficient for disruption.

4 Outcomes and next steps

Outcomes: An AI debate engine with replaceable components and a browser-based frontend.
We found no consistent gain in performance metrics from extra Hansard excerpts.

Next Steps: Expand testing with more providers and higher-performance language models.
Host the web browser platform on a server accessible through the internet.

Albert Sun — frontend, live chamber, initial CLI and Hansard grounding.
Kieran Joe — engine, architecture and supervisor coordination.

References

[1] Du et al. (2024). Multiagent Debate.
[2] Rauh & Schwalbach (2020). ParlSpeech V2.

Acknowledgements

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